FloodLAMP
Biotechnologies, PBC

Decentralized, Rapid Molecular
Disease Screening for $1

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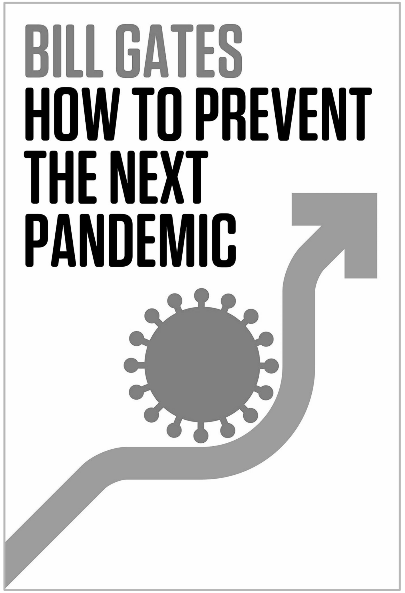
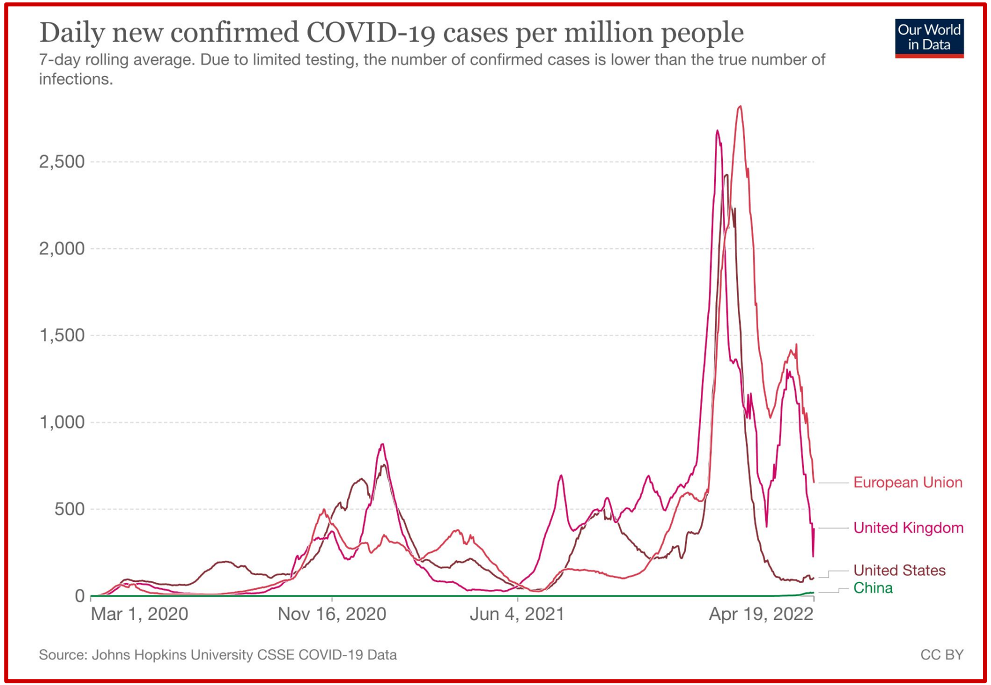
The Unknowable and How to Prepare for it



David Deutsch, “The Unknowable & how to prepare for it” at TEDxBrussels – (YouTube)

We *still* don't know how to prepare for another variant OR the next pandemic



Ideal Profile of Screening Tests

1 **Decentralized** — Local control, scalable during surge

2 **Rapid** — <1hr turn around time

3 **Molecular** — Greater accuracy compared to antigen tests
More quickly adaptable to new targets

4 **$1** — Low cost is critical for accessibility and wide distribution

[1] *Antigen tests catch only 23% of new asymptomatic infections same day as PCR ; 47% within 48 hrs [Comparison of Rapid Antigen Tests' Performance between Delta (B.1.61.7; AY.X) and Omicron (B.1.1.529; BA1) Variants of SARS-CoV-2: Secondary Analysis from a Serial Home Self-Testing Study]*

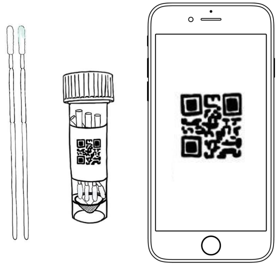
What This Looks Like

Programs operating in the real world, at the price of **$1.25** per person

A Disease Screening Program for Resiliency

1

At-home Accessioned Collection with Mobile App



Streamlines test processing

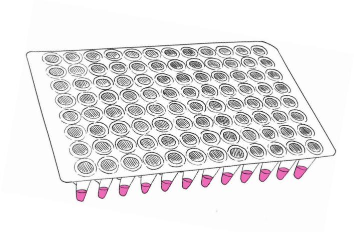
2

Family/ Household Pooling

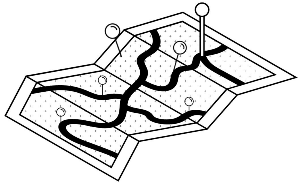
Extends protection

3

Rapid Molecular LAMP Test



<1hr True TAT



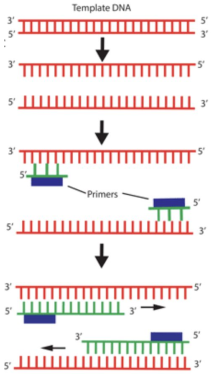
Run On or Near Site

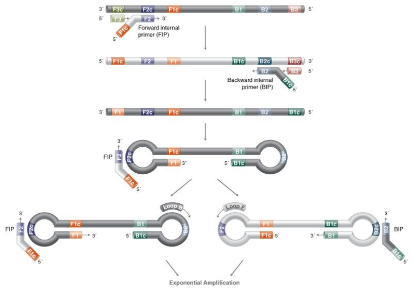
LAMP: Loop-Mediated Isothermal Amplification

- Alternative molecular amplification technology
- LAMP generates 10X more DNA product in 1/3 time compared to PCR
- Can be formulated for easy visual/camera-based readout
- Becoming an increasingly important part of diagnostics - many LAMP COVID EUAs!

Recommend NEB website and

global LAMP consortium review: (FloodLAMP co-author)
https://drive.google.com/file/d/1qdf-dL2cmbB7aK4rw57CgjPmWQ9K5X0F
FLOODLAMP ARCHIVE FILE PATH: various/glamp/gLAMP Global Consortium - JBT Review Paper (Sept 2021).md





	qPCR	LAMP
Equipment	qPCR machine (thermal cycler & optics)	simple heat block or water bath (isothermal)
Time	90 min	30 min
Capital Cost	$30K-100K	$300

Molecular Assay Atlas

PEOPLE

- Individual
- Pool: 2-20

SAMPLE

- Nasopharyngeal Swab (NP)
- Cheek Swab
- Anterior Nares Swab (AN)
- Saliva

INACTIV

- TCEP (Chem) + Heat
  Rabe Cepko, HUDSON
- Proteinase K (Enzyme) + Heat
  Saliva Direct
- Heat Only
- Commercial
  Zymo, Lucigen

PURIFIC

- Glass Milk
  Rabe Cepko
- Mag Beads
  Kellner, Yu
- Skip (Direct)
- Commercial
  Thermo, Zymo, Kingfisher, Qiagen

AMPLIF

- qPCR
- LAMP
  Colorimetric
  Fluorimetric
- Misc
  RPA, CRISPR, etc.

READOUT

- qPCR Machine
  Fluorescence
- Visual / Camera
  Color Change
- Plate Reader
  Absorbance
  Color Genomics
- Lateral Flow Strip

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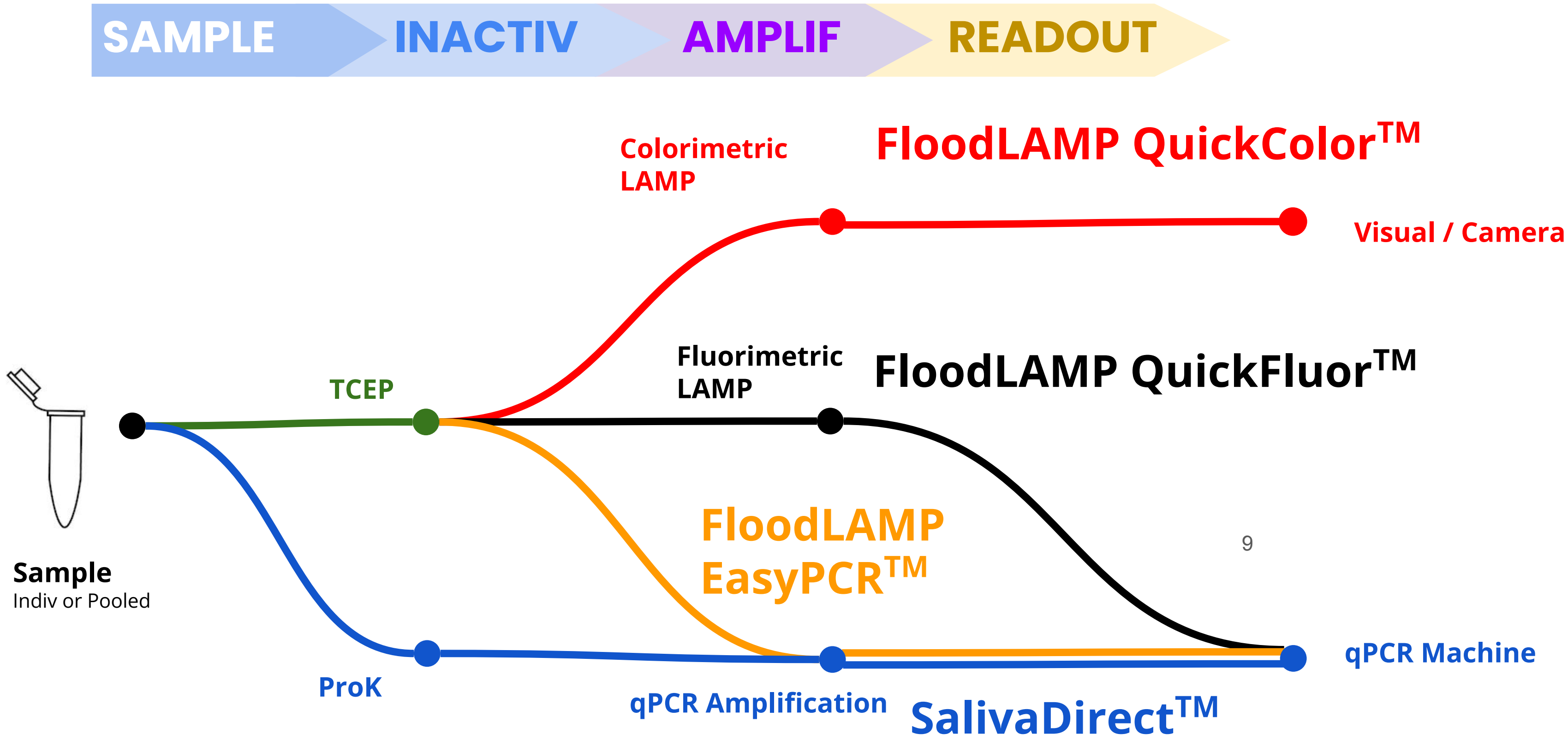
FloodLAMP Tests



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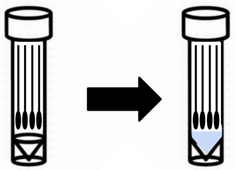
Direct PCR and LAMP Tests



Streamlined Sample Prep

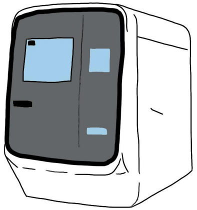
- Shelf stable TCEP/EDTA inactivation solution
- Added directly to dry swabs

- Same sample for both tests
- 2μL sample volume

visual readout

QuickColor™ LAMP Test

- Colorimetric LAMP reaction
- 3 primer sets mixed (AS1e, N2, E1)
- 25 minutes at 65°C

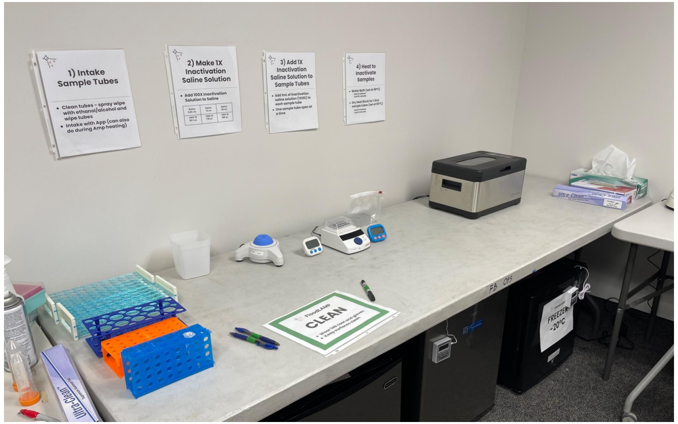


Standard RT-qPCR instrument

EasyPCR™ Test

- Multiple PCR Master Mixes Validated
- CDC primers in SalivaDirect config (N1, RNAseP)

Ideal for Austere Sites



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FloodLAMP Platform

Applications	COVID Response	Emerging Threats (Pandemic Prep)	TB Zika	Influenza STD's Cancer
Core Modules — PHYSICAL	Reagent Test Kits	Collection Kits	Standard Equipment	Standard Consumables
Core Modules — DIGITAL	App	Admin Portal	Training Program	Quality Management System
Network	EMS & Municipalities	Schools and Businesses	U.S. Federal Agencies (BARDA, ASPR, CDC, DoD, FEMA)	International Governments

Products

- All-in Test Kits
- Sample Processing Sites (Labs)
- Service, Support, Consulting
- Testing Program Management

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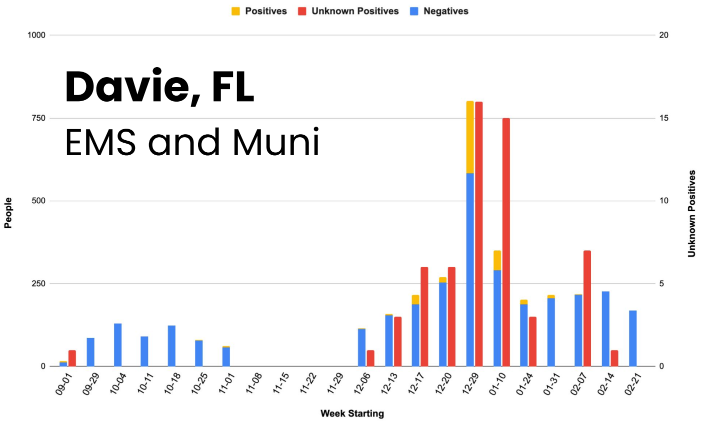
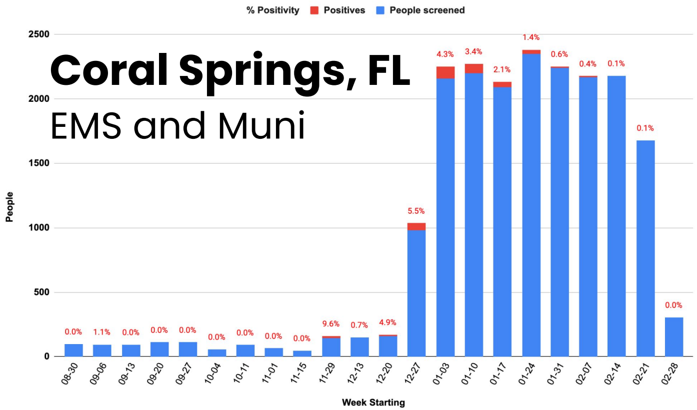
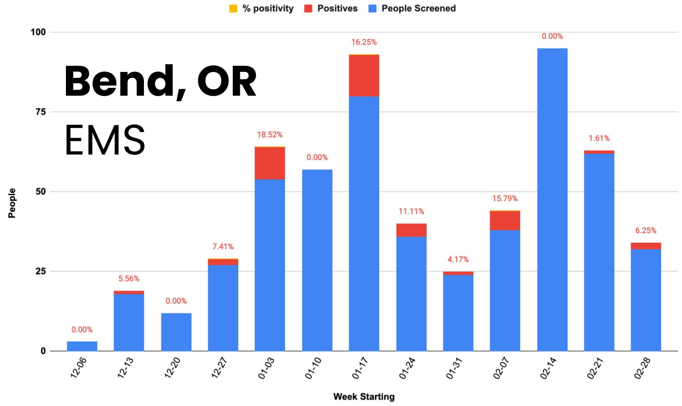
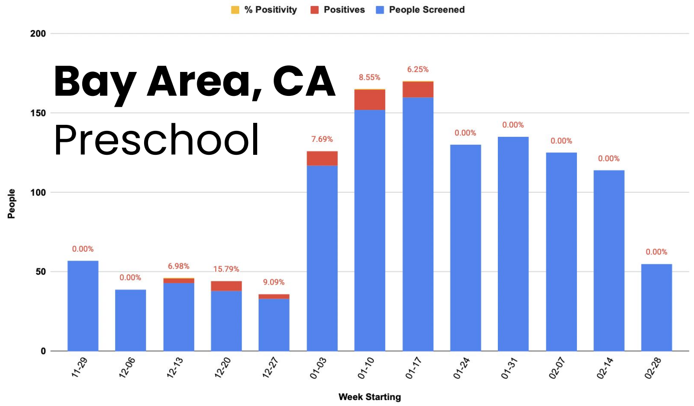
Scale Up During Omicron Surge









Real World Data

Comparison with diagnostic followup data in progress

- ➔ 22 FloodLAMP positives one day in late Dec and only 1 positive on BinaxNOW. **All** *22* became positive over next few days (reported).
- ➔ One site has 100% concordance of FloodLAMP positives with reflex PCR (34/34).

FloodLAMP Surveillance Testing Stats			**Thru Mar 2, 2022**
Org	Pools	People (Instances)	Positives (incl repeat indiv)
Staff, Preschool, Community	1,107	1,779	44
Conference	61	195	0
Summer Camp	190	696	1
Coral Springs, FL	7,400	21,676	348
Davie, FL	2,235	4,226	55
Bend Fire, OR	617	617	215
TOTAL	**11,610**	**29,189**	**663**

Known adverse results in >10K tests

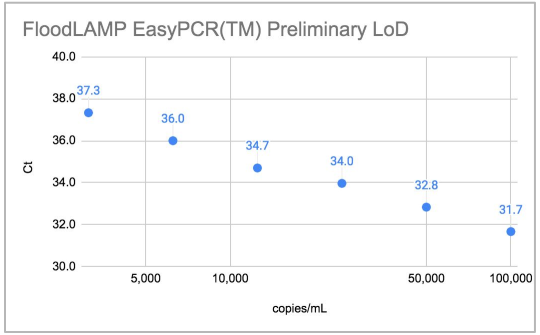
- ➔ 1 suspected false negative (lower instance than discordant lab PCR to FloodLAMP and antigen)
- ➔ 2 suspected false positives
- ➔ <1% to 5% rerun rate (initial inconclusives that rerun as negative)

Clinical Evaluation Data

Clinical evaluation performed by the Stanford CLIA Lab, with excellent results and praise on the **"really straightforward"** protocol.

EasyPCR™ Test

- 3 copies/µl LoD
- 98% sensitivity (PPA 39/40)
- 100% Specificity (40/40)
- No false positives



Gamma inactivated cell lysate from BEI spiked into raw clinical negative sample

FloodLAMP SwabDirect PCR Result	Comparator Positive	Comparator Negative	Total
Positive	39	0	39
Negative	1	40	41
Invalid	0	0	0
Total	40	40	80
Positive Agreement	97.5% (39/40) 95% CI: 86.8% to 99.9%		
Negative Agreement	100% (40/40) 95% CI: 91.2% to 100%		

QuickColor™ LAMP Test

- 12 copies/µl LoD
- 90% Sensitivity (PPA 36/40)
- Missed positives only high Ct (>36 with direct PCR)
- 100% Specificity (40/40)
- No false positives

FloodLAMP QuickColor Test Result	Comparator Positive	Comparator Negative	Total
Positive	36	0	36
Negative	4	40	44
Total	40	40	80
Positive Agreement	90.0% (36/40) 95% CI: 76.3% to 97.2%		
Negative Agreement	100% (40/40) 95% CI: 91.2% to 100%		

Source of Specimens: Stanford COVID-19 Clinical Testing Program
Specimen Type: Anterior Nares Swab in PBS, previously tested and frozen
Comparator Test: Hologic Panther Fusion SARS-CoV-2 Assay and Hologic Panther Aptima SARS-CoV-2 Assay

EUA Submissions – 2 Full + Pre EUA

FloodLAMP QuickColor™ COVID-19 Test

Instructions for Use v1.2

IVD

COVID-19 Emergency Use Authorization Only
For *in vitro* diagnostic (IVD) Use

www.floodlamp.bio
FloodLAMP Biotechnologies, PBC | 930 Brittan Ave. San Carlos, CA 94070 USA

FloodLAMP EasyPCR™ COVID-19 Test

Instructions for Use v1.1

IVD

COVID-19 Emergency Use Authorization Only
For *in vitro* diagnostic (IVD) Use

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FloodLAMP Pooled Swab Collection Kit DTC

For use with the FloodLAMP Mobile App

Open EUA - Disclosure of Components

In **contrast** to industry standard trade secret primer sequences and reaction components.

Table 1: Validated reagents used with the Test

Item	Concentration	Chemical Composition	Vendor	Catalog Number
TCEP	.5 M	tris(2-carboxyethyl)phosphine hydrochloride	Sigma-Aldrich Millipore Sigma	646547-10X1ML
EDTA	.5 M	Ethylenediaminetetraacetic acid	Thermo Fisher	15575020
NaOH	10 N	Sodium Hydroxide	Sigma-Aldrich	SX0607N-6
Nuclease-free Water		Ultrapure Water, nuclease-free	Thermo Fisher	10977015
NaCl	5 M	Sodium Chloride	Thermo Fisher	24740011
Guanidine HCl	6 M	Guanidine Hydrochloride	Sigma-Aldrich	SRE0066
Colorimetric LAMP MM*		Colorimetric LAMP Master Mix	New England Biolabs	M1804

* Item may not be substituted for equivalent. Only the specified vendor and catalog number may be utilized.

Table 2: Primer names and sequences

Primer Name	Sequence (5′-3′)
ORF1ab gene (AS1e)	
Orf1ab_FIP	TCAGCACACAAAGCCAAAAATTTATTTTTCTGTGCAAAGGAAATTAAGGAG
Orf1ab_BIP	TATTGGTGGAGCTAAACTTAAAGCCTTTTCTGTACAATCCCTTTGAGTG
Orf1ab_F3	CGGTGGACAAATTGTCAC
Orf1ab_B3	CTTCTCTGGATTTAACACACTT
Orf1ab_LF	TTACAAGCTTAAAGAATGTCTGAACACT
Orf1ab_LB	TTGAATTTAGGTGAAACATTTGTCACG
N Gene (N2)	
N2_FIP	TTCCGAAGAACGCTGAAGCGGAACTGATTACAAACATTGGCC
N2_BIP	CGCATTGGCATGGAAGTCACAATTTGATGGCACCTGTGTA
N2_F3	ACCAGGAACTAATCAGACAAG
N2_B3	GACTTGATCTTTGAAATTTGGATCT
N2_LF	GGGGGCAAATTGTGCAATTTG
N2_LB	CTTCGGGAACGTGGTTGACC

Table 7: Primer-Guanidine Solution

Component	Volume (1 reaction)	Volume (1 reaction x 100) 1 x 96-plate w/ 4% overage
10X LAMP Primer Mix	2.5 µL	250 µL
Guanidine HCl (400 mM)	2.5 µL	
Guanidine HCl (6 M)		16.7 µL
Nuclease-free Water	5.5 µL	783 µL
TOTAL VOLUME	**10.5 µL**	**1050 µL**

Table 8: Colorimetric LAMP Amplification Reaction

Component	Volume (1 reaction)	Volume (100 reactions)
Primer-Guanidine Solution	10.5 µL	1050 µL
Colorimetric LAMP MM	12.5 µL	1250 µL
SUBTOTAL VOLUME	**23 µL**	**2300 µL**
Sample	2 µL	
REACTION VOLUME	**25 µL**	

Open EUAs

Provides a path to **establishing generics in diagnostics**.
Enables **global dissemination** of highly respected FDA authorized tests.

	FDA IVD Typical IVD EUA	CDC EUA	Yale SalivaDirect™	ILLINOIS SHIELD	FloodLAMP EasyPCR™	FloodLAMP QuickColor™
Disclosure of all chemicals and reagents?	No	Yes	Yes	Yes	Yes	Yes
Chemical and reagents available from multiple vendors?	No	Yes	Yes	No	Yes	Yes
Disclosure of primer sequences?	No	Yes std for PCR	Yes	No ProptryThermo	Yes	Yes
Primers commercially available from multiple vendors?	No	Yes	Yes CDC Primers	No ProptryThermo	Yes CDC SD Primers	Yes Available but not launched
Supply chain robust?	No	No/Maybe	Yes	No/Maybe	Yes	Yes
EUA Sponsor Organization Type	For Profit Company	Govt	Academic Not for Profit	Academic Not/For Profit ?	Public Benefit Corp	Public Benefit Corp
Designation of CLIA labs	Kit Sales	N/A open RoR	Impact & Expansion	Impact & Expansion	Impact & Expansion	Impact & Expansion

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